But first you need to understand what it means, how it originated, and most importantly, the underpinning laws behind it. For that, we look at Conway's law to understand why we won't be able to change technology just on its own. We then look into the Team Topologies framework, which is, in essence, about socio-technical architecture. Finally, we look into cognitive load as one of the main ideas used by the Team Topologies framework.

1.6.1 Conway's law

Conway's law is defined as follows:

> Any organization that designs a system (defined broadly) will produce a design whose structure is a copy of the organization's communication structure.

Computer programmer Melvin Conway made this observation in a 1968 paper (http://mng.bz/Qv9j), just a decade after Fortran (the first widely used high-level language) was released. It didn't take us long to see this strong, almost gravitational force in practice. This *force* will sooner or later transform our architecture so it will look like our organizational structure. As architects, we need to be fully aware of the implied consequences.

We should take as an example the martial art Aikido. *Aiki* refers to the martial arts principle or tactic of blending with an attacker's movements to control their actions with minimal effort. The masters of Aikido teach that you should not put your force directly against the force of your opponent. Instead, you should take advantage of the force they are exerting. In the same way, we should be aware and take advantage of Conway's law.

Example: Central data warehouses

> Conway's law is a good explanation of why people say, "central data warehouse" but mean "central data warehouse operated by a central data team." It's because usually when you find a central data warehouse, it's been built by a central data team.

Conway's law simply tells us to take care of the organizational side of things and the technical side. Just changing technical things isn't going to have an